

1. Project Title

Global Technology and Internet Adoption Analysis (1990–2025): A World Development Indicators (WDI) Dashboard

2. Problem Statement

Technology and internet access are key drivers of economic growth, innovation, education, and social development. However, the level of internet adoption and digital infrastructure varies significantly across countries. This project analyzes global technology and internet adoption trends using the World Bank's World Development Indicators (WDI) dataset to identify patterns, compare countries, and provide insights that can support data-driven policy and investment decisions.

3. Dataset Description

The dataset used for this project was obtained from the **World Bank World Development Indicators (WDI)** database. It contains development statistics for over 200 countries and territories across multiple sectors, including economy, education, health, technology, trade, and environment.

For this analysis, the following technology-related indicators were selected:

- Individuals using the Internet (% of population)
- Mobile cellular subscriptions (per 100 people)
- Fixed broadband subscriptions (per 100 people)
- Secure Internet servers (per 1 million people)
- High-technology exports (% of manufactured exports)
- Population, total

The analysis covers data from **1990 to 2025**, depending on data availability.

4. Data Cleaning Process

The dataset was prepared using **Power Query** in Power BI through the following steps:

- Imported the World Development Indicators (WDI) CSV dataset.
- Filtered the dataset to retain only technology-related indicators.
- Removed unnecessary columns and irrelevant records.
- Unpivoted the yearly columns into a single **Year** column to simplify trend analysis.

- Renamed columns for consistency and readability.
- Converted data types appropriately (Year as Whole Number and Value as Decimal Number).
- Removed null and blank values where necessary.
- Created DAX measures for KPI calculations such as Average Internet Usage, Average High-Tech Exports, Average Secure Internet Servers, and Average Fixed Broadband Subscriptions.

5. Analysis Methodology

The analysis was performed using **Power BI Desktop**.

The dashboard includes:

- KPI Cards to summarize major performance indicators.
- Line Chart to analyze internet adoption trends over time.
- Scatter Plot to examine the relationship between internet usage and high-tech exports.
- Map Visualization to display global internet usage by country.
- Bar Charts showing the top countries in internet usage and mobile subscriptions.
- Interactive Slicers allowing users to filter by Year and Country.

Descriptive analytics techniques were applied to identify trends, compare countries, and explore relationships among technology indicators.

6. Key Findings

- Global internet usage has increased significantly from 1990 to 2025.
- Countries such as Kosovo, Liechtenstein, Monaco, and Iceland recorded some of the highest internet usage rates.
- Mobile phone subscriptions have grown rapidly across many countries, indicating increased digital connectivity.
- Countries with higher internet adoption generally show stronger high-technology export performance.
- Considerable disparities remain in technology adoption across different countries.

7. Insights

The analysis demonstrates that increased internet access is associated with stronger technological development and greater participation in the digital economy. While many countries have made

significant progress in expanding internet access, others continue to face challenges due to limited infrastructure and investment. The dashboard also shows that mobile technology has become the primary means of digital access in many regions, often outpacing fixed broadband adoption.

8. Recommendations

- Governments should invest more in broadband and digital infrastructure, especially in underserved areas.
- Policies that improve affordable internet access should be implemented to reduce the digital divide.
- Investment in research, innovation, and digital skills development should be encouraged.
- Countries should strengthen technology ecosystems to increase high-technology exports and economic competitiveness.
- Policymakers should use data-driven approaches to monitor digital development and guide future investments.

9. Conclusion

This project successfully analyzed global technology and internet adoption using the World Bank's World Development Indicators dataset. By applying data cleaning, transformation, visualization, and analytical techniques in Power BI, the project provided valuable insights into global digital development trends. The interactive dashboard enables stakeholders to explore internet adoption patterns, compare countries, and better understand the relationship between digital connectivity and technological advancement. The findings highlight the importance of continued investment in digital infrastructure to support sustainable economic growth and reduce inequalities in technology access.